

SIMIONESCU ROBERT-ALEXANDRU

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Education

Technical University of Cluj-Napoca

Computer Science – C.T.I

- 2nd Year Undergraduate Student

Cluj-Napoca, Romania

Oct. 2024 – Present

”Andrei Mureșanu” National College

Mathematics and Computer Science Specialization

- Baccalaureate Exam Grade: **9.00 / 10**

Bistrița, Romania

Sep. 2020 – Jul. 2024

Experience & Volunteering

Atomic Bots FIRST Tech Challenge Team

Co-founder / Programming branch

2022 – 2024

Bistrița, Romania

- Co-founded the Atomic Bots robotics team, coordinating the technical and programming efforts for the FIRST Tech Challenge competitions.
- Designed and implemented autonomous navigation and control algorithms for the robot using Java and Android Studio, integrating various sensors.
- Developed from scratch and optimized the team’s web interface, focusing on enhancing the user experience (UI/UX).

Projects

Atomic Bots Web Platform | React, Styled Components, JavaScript

Jan. 2026

- Designed and implemented the graphical interface (Frontend) for the robotics team, utilizing **React** and **Styled Components** for a modular architecture and responsive design.
- Optimized Single Page Application (SPA) navigation using **React Router** and prepared the application for efficient static distribution on the institution’s servers.

Leadership Coaching Platform | Python, Django, SQLite, HTML/CSS

Dec. 2025 - Jan. 2026

- Developed a full-stack web platform featuring a custom admin panel for independent blog article management by the client.
- Designed the relational database in **SQLite** and integrated a form-based contact system, utilizing the **SMTP** protocol for automated email confirmations.
- Managed the deployment process and continuous maintenance, including infrastructure administration via **cPanel** and **DNS** record routing to connect the custom domain with the **PythonAnywhere** environment.

IoT Air Quality Monitoring System | C++, Arduino UNO R4 WiFi

Dec. 2025

- Designed an integrated environment monitoring system using multiple sensors (DHT11, MQ-2, MQ-135) to detect temperature, air quality, and flammable gases.
- Developed a ”Gradient Smooth” linear interpolation mathematical algorithm for the fluid control of an RGB LED’s colors via **PWM** signals, visually reflecting thermal variations.
- Integrated the **NTP** protocol via WiFi for automatic time synchronization on an I2C LCD display, eliminating the need for RTC (Real-Time Clock) hardware.

Hardware VGA Controller | VHDL, FPGA (Nexys A7)

Apr. 2025 - May 2025

- Implemented a logic gate-level VGA controller on the **Nexys A7** FPGA board, capable of rendering geometric shapes in multiple colors controllable via hardware switches.
- Designed the control units (FSM) and counters to generate video synchronization signals (**H-Sync**, **V-Sync**).
- Developed an internal frequency divider to convert the 100 MHz clock signal to a stable 25 MHz *pixel clock*.

Nintendo Land | C#, .NET

Apr. 2023 - May 2023

- Developed a desktop application (virtual store) inspired by the Steam platform, including an authentication system and dynamic user profiles (avatars, XP leveling system).
- Implemented features for viewing reviews, media playback (trailers).

"U" Engine | *React, Next.js, Python, FastAPI, Google Gemini AI* Spring 2026

- Developed a full-stack tactical analysis web platform designed to aggregate and process Wyscout football data through a custom Python/FastAPI backend.
- Engineered an advanced AI module using the Google GenAI SDK (Gemini 2.5 Flash), implementing Dynamic Context Injection (opponent DNA, coach persona, raw stats) to generate structured JSON tactical strategies
- Designed an interactive React/Next.js frontend featuring dynamic pitch visualizations and statistical radars, while optimizing backend API latency and operational costs via a custom in-memory caching system.

32-bit MIPS Microprocessor | *VHDL,FPGA (Nexys A7), Xilinx Vivado, MIPS ASM* Mar. 2026 - May 2026

- Designed and implemented a fully functional 32-bit MIPS processor from scratch, featuring both Single-Cycle and 5-Stage Pipelined architectures within a unified repository.
- Engineered the complete datapath and control units, effectively managing Data and Control Hazards in the pipeline through static resolution strategies (compiler-level NOP insertion and branch delay slots) and inter-stage synchronization registers
- Validated the CPU hardware logic by writing and executing custom MIPS assembly algorithms (e.g., mathematical computations), simulated via Vivado testbenches and visualized on the FPGA's 7-segment display.

YouTube Downloader GUI | *C++, Qt Framework* Feb. 2026 - Mar. 2026

- Developing a robust desktop application (GUI) using **C++** and **Qt**, serving as an advanced visual interface for the *yt-dlp* utility.
- Implementing processing and downloading features for YouTube videos and playlists in multiple formats (MP4, MP3, FLAC).

Competitions

Coderun 2025 Cluj-Napoca / Nov. 2025

- Participated in 5 technical challenges testing our knowledge of React, Python/Django, C++/SDL3, Arduino, and cybersecurity.

Lovable HoHoHack 2025 Cluj-Napoca / Dec. 2025

- Collaborated in a team to build an online tutoring solution using Lovable development tools.
- **Link:** studdybuddyfullstackinut.lovable.app

"U" Hack Cluj-Napoca / Apr. 2026

- Competed in a fast-paced hackathon focused on developing innovative digital solutions for FC Universitatea Cluj.
- Built and pitched the "U Engine" prototype, successfully integrating complex Wyscout data with AI to provide the technical staff with instant, actionable tactical strategies.

Technical Skills

Programming Languages: C, C++, C#, Java, Python, SQL, HTML/CSS, JavaScript, TypeScript, VHDL, MySQL
Frameworks & Libraries: React, Django, Arduino, SDL3, Java Swing, .NET, Qt, FastAPI
Tools: Git, GitHub, Fedora Linux, Visual Studio Code, IntelliJ, Android Studio, CLion, DataGrip
Languages:Romanian(Native), English (C1 CAE), German (A1), French (B1), Hungarian(A2)

Courses and Certifications

Google Cybersecurity Seminars @ Universitatea tehnica din Cluj-Napoca Spring 2026

Database Foundation & Programming with SQL @ ORACLE Academy Feb. 2026

Certified Integrative NLP Practicioner @ NLP Integrativ Apr. 2026

English: Advanced (C1), Cambridge English: Advanced (CAE) – Score 192 Nov. 2023